GOVERNMENT & POLICY

# How your solar rooftop became a national security issue

Connie Loizos — 1:58 PM PDT · August 15, 2025

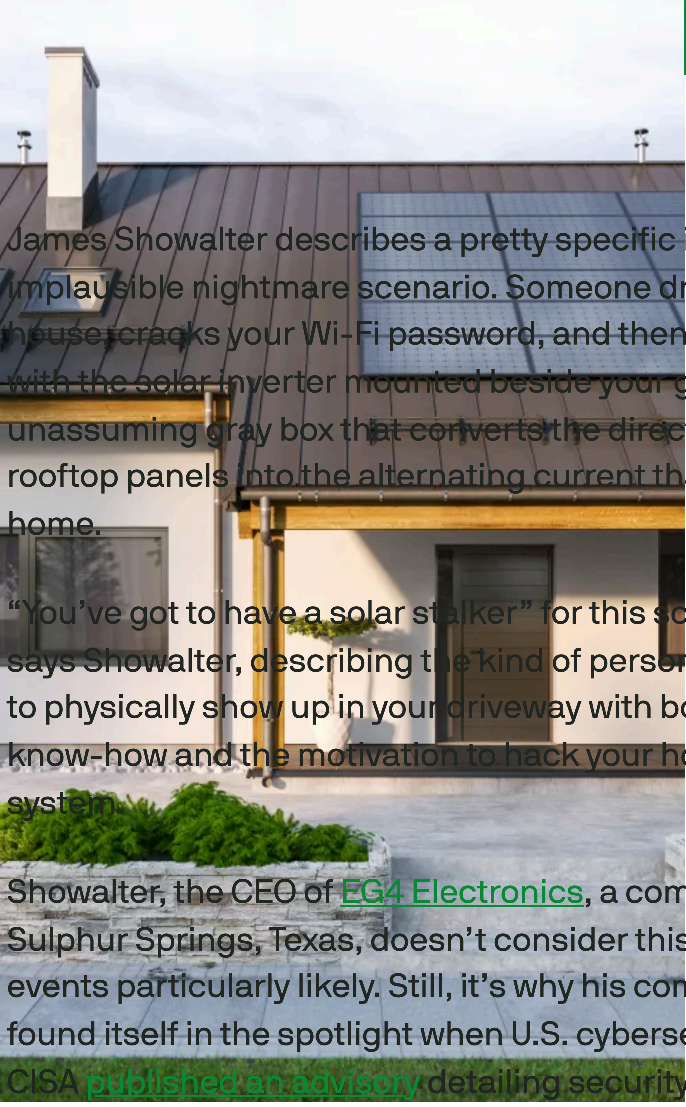

IMAGE CREDITS: IMAGINIMA / GETTY IMAGES

James Showalter describes a pretty specific if not entirely implausible nightmare scenario. Someone drives up to your house, cracks your Wi-Fi password, and then starts messing with the solar inverter mounted beside your garage — that unassuming gray box that converts the direct current from your rooftop panels into the alternating current that powers your home.

"You've got to have a solar stalker" for this scenario to play out, says Showalter, describing the kind of person who would need to physically show up in your driveway with both the technical know-how and the motivation to hack your home energy system.

Showalter, the CEO of EG4 Electronics, a company based in Sulphur Springs, Texas, doesn't consider this sequence of events particularly likely. Still, it's why his company last week found itself in the spotlight when U.S. cybersecurity agency CISA published an advisory detailing security vulnerabilities in EG4's solar inverters. The flaws, CISA noted, could allow an attacker with access to the same network as an affected inverter and its serial number to intercept data, install malicious firmware, or seize control of the whole system.

For the roughly 55,000 customers who own EG4's affected inverter model, the episode probably felt like an unsettling introduction to a device that they little understand. What they're learning is that modern solar inverters aren't simple power converters anymore. They now serve as the backbone of home energy installations, monitoring performance, communicating with utility companies, and, when there's excess power, feeding it back into the grid.

Much of this has happened without people noticing. "Nobody knew what the hell a solar inverter was five years ago," observes Justin Pascale, a principal consultant at Dragos, a cybersecurity firm that specializes in industrial systems. "Now we're talking about it at the national and international level."

## Security shortcomings and customers' complaints



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Some of the numbers highlight the degree to which individual homes in the U.S. are becoming miniature power plants. According to the U.S. Energy Information Administration, small-scale solar installations — primarily residential — grew more than fivefold between 2014 and 2022. What was once the province of climate advocates and early adopters became more mainstream owing to falling costs, government incentives, and a growing awareness of climate change.

Each solar installation adds another node to an expanding network of interconnected devices, each one contributing to energy independence but also becoming a potential entry point for someone with malicious intent.



When pressed about his company's security standards, Showalter acknowledges its shortcomings, but he also deflects. "This is not an EG4 problem," he says. "This is an industry-wide problem." Over a Zoom call and later, in this editor's inbox, he produced a 14-page report cataloguing 88 solar energy vulnerability disclosures across commercial and residential applications since 2019.

Not all of his customers — some of whom took to Reddit to complain — are sympathetic, particularly given that CISA's advisory revealed fundamental design flaws: communication between monitoring applications and inverters that occurred in unencrypted plain text, firmware updates that lacked integrity checks, and rudimentary authentication procedures.

"These were fundamental security lapses," says one customer of the company, who asked to speak anonymously. "Adding


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insult to injury," continues this individual, "EG4 didn't even bother to notify me or offer suggested mitigations."

Asked why EG4 didn't alert customers straightaway when CISA reached out to the company, Showalter calls it a "live and learn" moment.

"Because we're so close [to addressing CISA's concerns] and it's such a positive relationship with CISA, we were going to get to the 'done' button, and then advise people, so we're not in the middle of the cake being baked," says Showalter.

TechCrunch reached out to CISA earlier this week for more information; the agency has not responded. In its advisory about EG4, CISA states that "no known public exploitation specifically targeting these vulnerabilities has been reported to CISA at this time."

## Connections to China spark security concerns

While unrelated, the timing of EG4's public relations crisis coincides with broader anxieties about the supply chain security of renewable energy equipment.

Earlier this year, U.S. energy officials reportedly began reassessing risks posed by devices made in China after discovering unexplained communication equipment inside some inverters and batteries. According to a Reuters investigation, undocumented cellular radios and other communication devices were found in equipment from multiple Chinese suppliers — components that hadn't appeared on official hardware lists.

This reported discovery carries particular weight given China's dominance in solar manufacturing. That same Reuters story noted that Huawei is the world's largest supplier of inverters, accounting for 29% of shipments globally in 2022, followed by Chinese peers Sungrow and Ginlong Solis. Some 200 GW of European solar power capacity is linked to inverters made in China, which is roughly equivalent to more than 200 nuclear power plants.

The geopolitical implications haven't escaped notice. Lithuania last year passed a law blocking remote Chinese access to solar, wind, and battery installations above 100 kilowatts, effectively restricting the use of Chinese inverters. Showalter says his company is responding to customer concerns by similarly starting to move away from Chinese suppliers and toward components made by companies elsewhere, including in Germany.

But the vulnerabilities CISA described in EG4's systems raise questions that extend beyond any single company's practices or where it sources its components. The U.S. standards agency NIST warns that "if you remotely control a large enough number of home solar inverters, and do something nefarious at once, that could have catastrophic implications to the grid for a prolonged period of time."

The good news (if there is any), is that while theoretically possible, this scenario faces a lot of practical limitations.

Pascale, who works with utility-scale solar installations, notes that residential inverters serve primarily two functions: converting power from direct to alternating current, and facilitating the connection back to the grid. A mass attack would require compromising vast numbers of individual homes simultaneously. (Such attacks are not impossible but are more likely to involve targeting the manufacturers themselves, some of which have remote access to their customers' solar inverters, as evidenced by security researchers last year.)

The regulatory framework that governs larger installations does not right now extend to residential systems. The North American Electric Reliability Corporation's Critical Infrastructure Protection standards currently apply only to larger facilities producing 75 megawatts or more, like solar farms.

Because residential installations fall so far below these thresholds, they operate in a regulatory gray zone where cybersecurity standards remain suggestions rather than requirements.

But the end result is that the security of thousands of small installations depends largely on the discretion of individual manufacturers that are operating in a regulatory vacuum.

On the issue of unencrypted data transmission, for example, which is one reason EG4 received that slap on the hand from CISA, Pascale notes that in utility-scale operational environments, plain text transmission is common and sometimes encouraged for network-monitoring purposes.

“When you look at encryption in an enterprise environment, it is not allowed,” he explains. “But when you look at an operational environment, most things are transmitted in plain text.”

Put another way, the real concern isn’t an immediate threat to individual homeowners. Instead it ties to the aggregate vulnerability of a rapidly expanding network. As the energy grid becomes increasingly distributed, with power flowing from millions of small sources rather than dozens of large ones, the attack surface expands exponentially. Each inverter represents a potential pressure point in a system that was never designed to accommodate this level of complexity.

Showalter has embraced CISA’s intervention as what he calls a “trust upgrade” — an opportunity to differentiate his company in a crowded market. He says that since June, EG4 has worked with the agency to address the identified vulnerabilities, reducing an initial list of 10 concerns to three remaining items that the company expects to resolve by October. The process has involved updating firmware transmission protocols, implementing additional identity verification for technical support calls, and redesigning authentication procedures.

But for those like the anonymous EG4 customer who spoke with frustration about the company’s response, the episode highlights the odd position that solar adopters find themselves in. They purchased what they understood to be climate-friendly tech, only to discover they’d become unwitting participants in a knotty cybersecurity landscape that few seem to fully comprehend.

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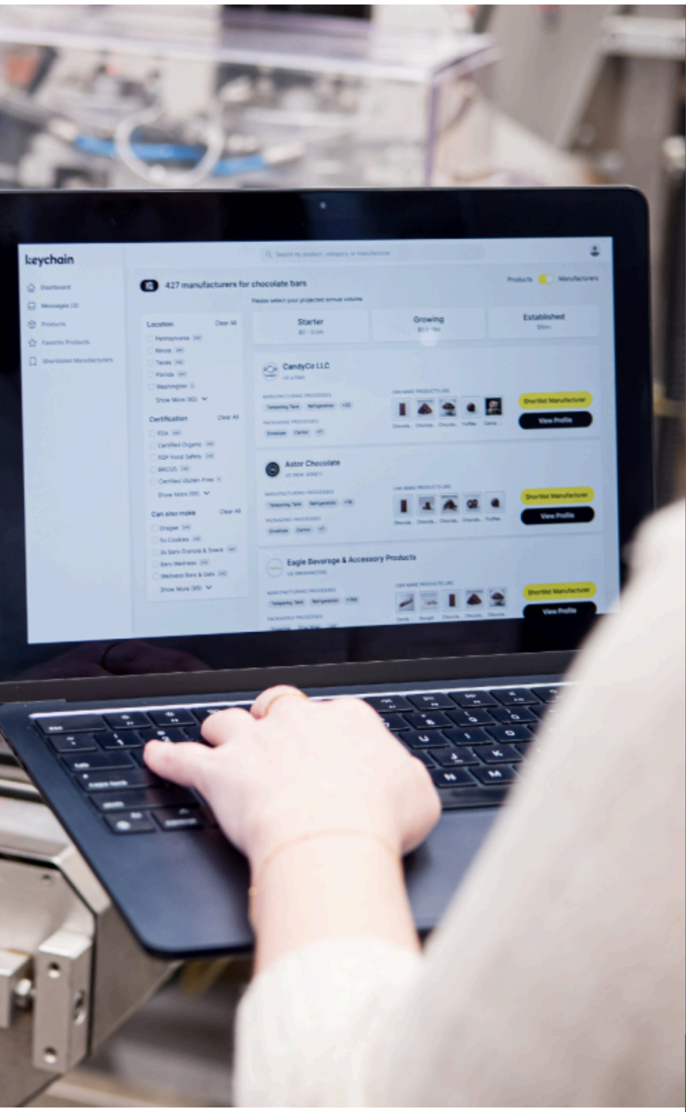


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ENTERPRISE

# CPG startup Keychain snags $30M to build in India, grow in the U.S.

Jagmeet Singh — 7:00 AM PDT · August 19, 2025

Keychain, a U.S. startup that helps consumer brands find manufacturing partners, has raised $30 million in fresh funding as it looks to scale its India-based development team to drive growth in North America.

While headquartered in New York, Keychain operates as a distributed company with its core engineering and product development centered in India. The startup is doubling down on this model with the new funding, aiming to grow its engineering, product design, and analytics teams in Gurugram from 35 to 70 in the coming months, and to around 100 within a year. This India-based team already represents half of Keychain's 70-person global headcount, with about 20 employees in New York and the remainder in Austin, handling partnerships, go-to-market, and sales.

The strategy is deliberate. Despite only serving Western markets, Keychain has built its primary development operations in in Gurugram — which is the country's second-largest tech hub after Bengaluru — to develop its consumer packaged goods (CPG) platform for clients in North America. The software platform already helps eight of the top 10 retailers, including 7-Eleven and Whole Foods, and seven of the top CPG brands, such as General Mills, connect with potentially suitable manufacturers, according to the startup. So why India?

"It's the talent, depth, availability, and the speed with which you can access talent of that depth and availability [in India]," said Oisin Hanrahan, co-founder and CEO of Keychain, in an interview.

Hanrahan co-founded Keychain in 2023 with Umang Dua — his co-founder at Handy, a home services software startup later acquired by Angi — and Jordan Weitz. Dua, who is originally from New Delhi, has been a "natural advantage" in building Keychain's core teams in Gurugram, Hanrahan said.

KEYCHAIN CO-FOUNDERS JORDAN WEITZ, OISIN HANRAHAN, AND UMANG DUA (LEFT TO RIGHT)

IMAGE CREDITS:KEYCHAIN

Both Hanrahan and Dua spent time structuring Keychain's teams across India and the U.S., ultimately choosing India as the company's engineering hub. The decision was shaped by their experience at Handy and Angi, where they found it challenging to build a "sustainable, enduring" engineering team in the U.S.

"We've thought about engineering as: how do we build a core, sustainable engineering organization that can get to scale reasonably quickly, that has endurance, that's got deep talent pools, and AI exposure that can take on real, important challenges, that's commercially minded? And we looked at where we had those teams before, when we were at Handy and Angi, and obviously, India is just an amazing location and really checks a lot of those boxes," Hanrahan told TechCrunch.



Several U.S. startups, especially those developing SaaS solutions, base their engineering and product teams in cities like Bengaluru, Gurugram, and Noida. In recent months, the country has also seen a wave of multinational companies establishing offshore hubs, often referred to as global capability centers. But unlike most of these firms — which also target Indian consumers even as many say India is harder to sell into — Keychain stands apart. It more closely resembles companies like the UK's Deliveroo and Southeast Asia's Gojek and Grab — all of which tap into India's tech talent for product development and R&D without having a market presence in the country.

"India's position as a global technology hub has made it a

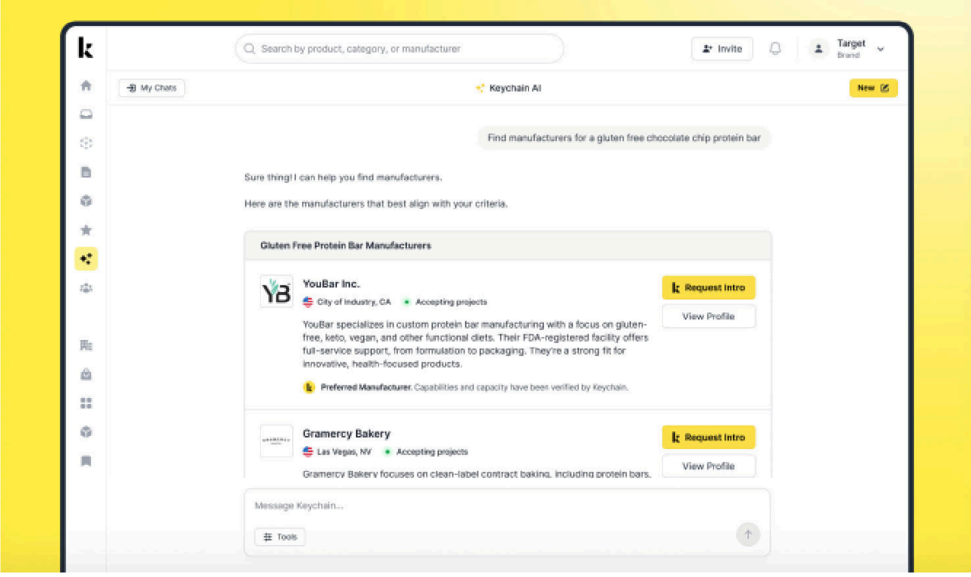
20,000 brands and retailers to find manufacturing partners — but also to build new AI-powered software that helps manufacturers manage their product cycles more efficiently and with better oversight.

Called KeychainOS, the software will have four modules, with the first one already available. This module helps manufacturers comply with their food safety requirements, using AI to take quantitative data and convert it into a qualitative report that can be shared with auditors. The software can also pull data using natural language when an auditor requests a specific insight, Hanrahan told TechCrunch.

The other three modules of the software will focus on purchasing and procurement, inventory, and production planning, the executive noted.

The OS offering will compete with traditional ERP systems like Oracle, QAD, and Plex, which require add-ons like TraceGains and Redzone to be usable for manufacturers, the startup said.

In addition to its KeychainOS for manufacturers, Keychain has embedded AI into its search and discovery layer to help retailers quickly find relevant third-party manufacturers for their products.




KEYCHAIN'S SEARCH AND DISCOVERY LAYER WITH AN AI INTEGRATION

IMAGE CREDITS:KEYCHAIN

Keychain already helps brands and retailers find third-party manufacturers in food, beverage, supplements, health, and beauty categories and is looking to expand its platform to pet and household products later this year.

Currently, the startup serves businesses in the U.S. and Canada and is aiming to enter Europe later this year.

While the startup offers its software free to brands and retailers, manufacturers pay to access the platform and get discovered. KeychainOS provides them with another reason to engage.

Keychain already has over 30,000 manufacturers on its platform, with "hundreds and hundreds" paying to use it. These customers pay anywhere from $10,000 to over $100,000, Hanrahan said, adding that the startup earns around $20,000 per manufacturer annually on average.

Keychain's Series B round was led by Wellington Management and existing investor BoxGroup, alongside other existing investors. With this funding, the startup has raised $68 million in total. Of that, Hanrahan told TechCrunch that the startup still has over $50 million in the bank.

The startup had a post-money valuation of $260 million in its last round of $15 million in November 2024. Hanrahan did not disclose the current valuation but said it was a "good step up."

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# TikTok's latest feature lets college students find and connect with classmates

Aisha Malik — 6:00 AM PDT · August 19, 2025

In a move reminiscent of Facebook's early days, TikTok is launching a new feature that allows college students to find and connect with others on their campus. The feature, called Campus Verification, lets users add their college campus to their TikTok profile and browse a list of students at their school.

To access the feature, users need to navigate to their profile and tap the "Add school" button. From there, they need to enter their school name and then select their graduation year.

To verify that they actually go to the school they entered, they need to provide their academic email address. Once complete, their college name and graduation year will be displayed on their profile.

Users can then browse their school's page, filter by graduation year, and view all of the users who have added the school to their profile. They can also sort the users list to show the most-followed users first.

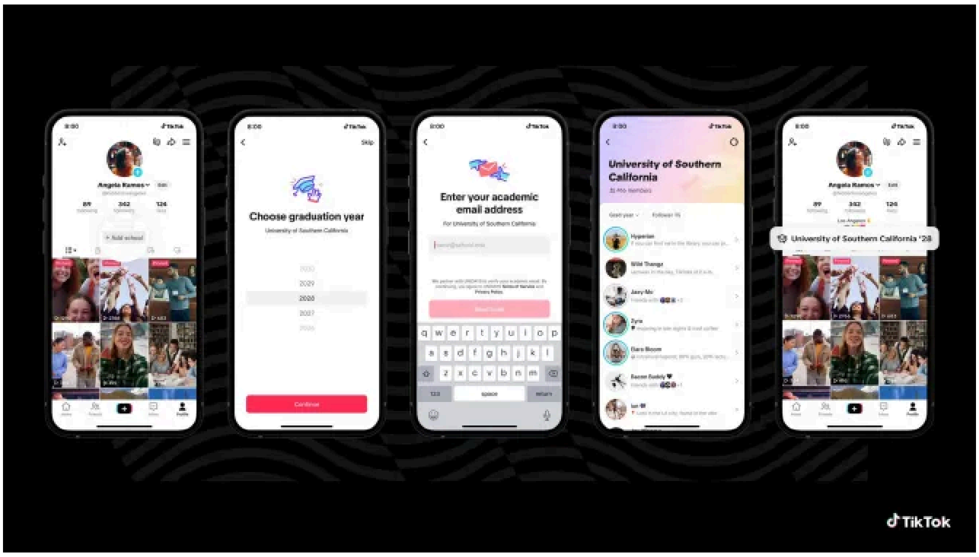


IMAGE CREDITS:TIKTOK

The feature is available across more than 6,000 universities through TikTok's partnership with UNiDAYS, a student verification platform that confirms student status.

TikTok says the idea behind the feature is to turn its platform into a hub for real-world connections and belonging on campus, which is essentially what Facebook's original mission was when it launched in 2004 as a way for Harvard students to connect with one another.

TikTok's new feature could be a welcome addition for college students looking to connect with others on campus. However, it also raises concerns about digital privacy, as it could make it easier for users to track others online. Of course, the feature is optional, so those who prefer to keep their TikTok presence separate from their student life can choose to simply not add their school to their profile.

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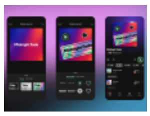

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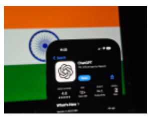

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It's worth noting that the launch comes as Instagram was spotted testing a similar feature last year that would allow users to display their school info and graduation year on their profile and connect with other students. It's unknown if the feature was scrapped or if the Meta-owned platform still plans to roll it out.

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Aisha is a consumer news reporter at TechCrunch. Prior to joining the publication in 2021, she was a telecom reporter at MobileSyrup. Aisha...

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